I. Results and Discussion

This chapter presents the experimental results of the proposed FeatureSAGE framework and compares its performance with the baseline SAGE model on few-shot image generation tasks. The experiments are carried out on two datasets, FUNIT Animal Faces and 102 Flowers, with a 1-shot image generation task to evaluate the ability of the proposed framework when faced with a limited amount of data. In order to evaluate the performance in high and low resolution settings, the experiments are conducted at 1024x1024 resolution only. The quality of the generated images and their perceptual similarity to real images are measured using Fréchet Inception Distance (FID) and Learned Perceptual Image Patch Similarity (LPIPS), respectively. A comparison between the SAGE model with a pSp inverter and the FeatureSAGE framework with StyleFeatureEditor and feature editing is expected to shed light on the effect of improved latent inversion on image quality and editability.

A. Quantitative Results

1) Quantitative Comparison Results between SFSAGE and Baselines:

B. Analysis of Quantitative Results

This section analyzes the quantitative performance trends observed in Table I, focusing on how the proposed SFSAGE model compares against AGE and SAGE baselines across the Flowers and Animal Faces datasets at 1024×1024 resolution only using FID and LPIPS metrics.

1) 102 Flowers: On a high resolution setting (1024×1024), SFSAGE obtains an FID of 45.60, which corresponds to a 15.8% improvement over SAGE (54.17) and a 25.9% improvement over AGE (61.57). This substantial FID reduction demonstrates improved distributional alignment with the real flower images in the Inception feature space. In terms of perceptual diversity, SFSAGE achieves an LPIPS of 0.4209, compared to SAGE (0.4528) and AGE (0.4108). This represents a 7.0% decrease relative to SAGE and a 2.5% increase relative to AGE. The LPIPS values across all three methods range from 0.4108 to 0.4528, indicating that perceptual diversity remains relatively consistent across the approaches. The results show that SFSAGE achieves FID improvements of 15.8-25.9% while maintaining LPIPS within 7.0% of the baseline methods.

2) Animal Faces: On the Animal Faces HQ dataset at 1024×1024, SFSAGE achieved an FID of 49.60, outperforming SAGE (60.92) by 18.6% and AGE (62.13) by 20.2%. This reduction in FID demonstrates SFSAGE's superior distributional alignment with the real data distribution in the Inception feature space. In terms of perceptual diversity, SFSAGE achieves an LPIPS of 0.4811, compared to SAGE (0.4867) and AGE (0.5033). This represents a 1.2% decrease relative to SAGE and a 4.4% decrease relative to AGE. The LPIPS differences across all three methods are relatively small, with values ranging from 0.4811 to 0.5033, indicating comparable levels of perceptual diversity in the VGG feature space. The results show that SFSAGE achieves substantial FID improvements (18.6-20.2%) while maintaining LPIPS values within 4.4% of the baseline methods.

C. Qualitative Evaluation

1) Visual Comparison Across Models:

2) Visual Observations:

a) Flowers: Figures 1 and 2 describe the qualitative analysis of the results of the edited flowers based on the baseline models AGE and SAGE, and the proposed model SFSAGE. The models are compared with regard to their ability to maintain the colors properly, the sharpness of the images, the level of contrast, and realism. As shown in Figure 1, the proposed model performs well in the retention of the pink petal colors from the input images. It also performs much better based on the level of semantic consistency. Note that the proposed model also fails to represent the combined red and yellow colors of the pistil from the input images. On the other hand, the proposed model performs much better than both the input images in the sharpness of the petal. Although there are inconsistencies from the input images, the proposed model also performs much better. In addition, the level of contrast in the proposed model results in the sharpest flower among all the models.

Further, the advantages of SFSAGE are highlighted by Figure 2. The sharpness in petal edges is better retained by the model compared to AGE and SAGE, which allows for a more photo-realistic and organic rendering. More varied outputs have also resulted from SFSAGE; for instance, in one generated image, the flower is presented with a slight tilt, reflecting strong attribute manipulation. The model maintains higher clarity, with balanced contrast and natural color distribution, while AGE slightly looks pale with reduced contrast in areas, which is somewhat settled by SAGE as it balances the contrast. Overall, the 65 generated flower images produced by SFSAGE are the sharpest, most realistic, and visually consistent as compared to the other models being tested.

b) Animal Faces: Figure 3 provides a comparison of the outputs produced by the AGE, SAGE, and proposed SFSAGE model on the Animal Faces data set using a picture of a dog as input. As observed, the outputs produced by the models have managed to maintain the color and texture of the input data, implying success by these models in maintaining basic characteristics produced by each model. Considering the clarity and image properties, all the outputs produced by these models can be considered similar, since there is minimal blurring effect and texture preservation on the fur of the input image. Consistency and overall properties, on the other hand, differ significantly, since there is more diversity apparent in the outputs produced by the AGE and SAGE models, with a slight difference in orientation between the two outcomes produced by each model. The more consistent results produced by the proposed SFSAGE model, on the contrary, demonstrate success towards maintaining integrity while providing semantic edits. Also, there are some minor enhancements in the realism of the output, whereby the output from SFSAGE shows slightly more coherent shading and blending of the

Resolution	Model	Flowers		Animal Faces	
		FID ↓	LPIPS ↑	FID ↓	LPIPS ↑
1024	AGE (Baseline)	61.57	0.4108	62.13	0.5033
1024	SAGE (Baseline)	54.17	0.4528	60.92	0.4867
1024	**SFSAGE (Ours)**	45.60	0.4209	49.60	0.4811

TABLE I
QUANTITATIVE COMPARISON OF AGE, SAGE, AND THE PROPOSED SFSAGE ACROSS FLOWERS AND ANIMAL FACES DATASETS AT 1024×1024 RESOLUTION. LOWER FID AND HIGHER LPIPS VALUES INDICATE BETTER PERFORMANCE.

Fig. 1. Comparison across AGE, SAGE, and the proposed SFSAGE model under one-shot setting, where a single reference image is used to condition the model during inference.

Fig. 2. Comparison across AGE, SAGE, and the proposed SFSAGE model under one-shot setting with a different image input.

Fig. 3. Comparison across AGE, SAGE, and the proposed SFSAGE model on the Animal Faces dataset under one-shot setting. Each model generates two outputs per input image to demonstrate edit diversity and semantic consistency.

fur patterns. This indicates that although all these models demonstrate the ability to sustain basic attributes like color and texture, the SFSAGE model has the best consistency in output.

Figure 4, the similar behavior to the third image input for SFSAGE is observed, where it maintains consistency in terms of pose within the results, showing that it does not have great variability compared to AGE and SAGE, which have more diverse orientations. At the same time, it may be noted that SFSAGE is better at maintaining consistency of shape and structure for eyes, thus improving realism. It is also observed that the results from SFSAGE have better clarity and contrast, improving image quality, particularly for fur texture and shading. On the other hand, it may be noted that AGE performs best in terms of diversity among all models, particularly for pose variation. Yet, it is observed that this diversity may also decrease image clarity.

D. Failure Cases and Limitations

During experimentation and evaluation of the FeatureSAGE framework, several limitations and failure cases were observed:

1) Built using a pretrained StyleGAN2 generator, FeatureSAGE requires fine-tuning on specific data to enable accurate image reconstruction and generation. Because of this setup, its use remains limited to areas where such pre-trained models are already available. When the base model lacks diversity - say, missing certain categories - those gaps carry forward into results. Oddly enough, flaws embedded in the original network's hidden representations often surface during editing tasks. Depending on training history, these quirks may distort outcomes or skew how traits appear across faces. As a result, changes made through the system can reflect old imbalances instead of producing balanced updates.
2) Computation Overhead. Adding the StyleFeatureEditor

Fig. 4. Comparison across AGE, SAGE, and the proposed SFSAGE model under one-shot setting with a different dog input image. Each model generates two outputs per input to illustrate diversity and semantic consistency.

inverter along with the Feature Editor demands more processing than the base SAGE setup. These parts boost reconstruction quality yet require greater resources. Performance gains come at a cost in speed. Though results improve, the system now takes longer per operation. Efficiency drops slightly due to added layers. Each forward pass grows heavier. More complex steps slow down inference. Trade-offs appear in timing versus output fidelity. Higher precision arrives with increased load. Processing time rises alongside capability. System throughput narrows somewhat. Resource demand climbs without drastic changes elsewhere.

Due to higher demands on processing resources, these features take up more GPU memory while slowing down inference. As a result, performance can drop when handling big datasets or time-sensitive tasks. Focusing on different scales of attributes could open new paths forward. Still, adjusting how strongly edits are applied matters just as much. One path involves testing models other than StyleGAN2, which might ease reliance on a single framework. Generalization may get stronger when diverse designs enter the picture. Exploring such options becomes essential given current constraints.

II. Conclusion

A. Summary of Findings

1) Summary:

a) 1. Integration of StyleFeatureEditor (SFE) Encoder: FeatureSAGE integrates the StyleFeatureEditor Encoder into the original SAGE framework, replacing the pSp encoder to enable improved latent representations that better capture distributional characteristics of the data. This integration resulted in an FID of 45.60 on the 102 Flowers dataset and 49.60 on the Animal Faces dataset at 1024×1024 resolution, representing improvements of 15.8% and 18.6% over SAGE, and 25.9% and 20.2% over AGE, respectively.

b) 2. Integration of Feature Editor: The Feature Editor module allows fine-grained and semantically consistent modifications by operating directly on intermediate feature representations, enhancing the expressiveness of attribute edits. Its inclusion contributed to sharper outputs with better color fidelity and structural coherence, as evidenced by SFSAGE's consistent qualitative advantages over both AGE and SAGE across Flowers and Animal Faces datasets.

c) 3. Resolution-Dependent and Dataset-Adaptive Editing: FeatureSAGE adapts its editing strategies according to both image resolution and dataset characteristics, ensuring appropriate trade-offs between edit diversity, stability, and realism. On the Flowers dataset, SFSAGE maintained an LPIPS of 0.4209, within 7.0% of SAGE (0.4528), while achieving superior FID. On Animal Faces, an LPIPS of 0.4811 remained within 4.4% of the baseline methods, confirming that quality improvements did not significantly sacrifice perceptual diversity.

d) 4. Comprehensive Benchmarking Against Baselines: The framework demonstrates robust performance across structurally diverse datasets and provides a generalizable approach to few-shot image generation. SFSAGE outperformed both AGE and SAGE on all FID metrics across both datasets, while maintaining comparable LPIPS scores within a range of 0.4108 to 0.4528 on Flowers and 0.4811 to 0.5033 on Animal Faces, confirming that distribution alignment improved without compromising output diversity.

B. Limitations

Though FeatureSAGE holds promise in attribute group editing and limited example image synthesis, some constraints observed during testing indicate that there is scope for further research. The first is the reliance on a pre-trained StyleGAN2 generator, which restricts the application of this tool only to domains where such generators have been developed, and any biases in the model may have been passed on to the output. The second is the potential performance impact of integrating the StyleFeatureEditor with the Feature Editor, which increases computational requirements. Future research could focus on other base models, incorporate a multi-level feature approach to improve consistency between edits, optimize computation for more general or faster models, and extend few-shot adaptation to work well even on entirely new classes.

C. future Work

1) Efficiency Optimizations:

2) Broader Applications:

D. Conclusion